JOINT ADMISSION TEST FOR MASTERS • IIT JAM MATHEMATICS (MA)

Finite Dimensional Vector Spaces (2022–2026)

Category: Linear Algebra and Algebra • Official Mock Test Series

TOTAL QUESTIONS	TOTAL MARKS	TEST DURATION	PAPER PATTERN
28	46 M	84 Min	MCQ • MSQ • NAT (2022–20

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	12	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	7	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	9	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

1. **Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
2. **Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
3. **Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
4. **Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
5. **Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
6. **Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at **vaibhavgeometer.github.io**.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2026 • Q40 • ID: JAM_2026_Q40) MSQ +2 M Neg: Nil (0)

Q.40	Let V be the subset of $\mathbb{R}$ defined by $$V = \left\{\frac{a + b\sqrt{2}}{c + d\sqrt{2}} : a, b, c, d \in \mathbb{Q}, c^2 + d^2 \neq 0\right\}.$$ Which of the following statements is/are FALSE?
(A)	V is closed under the usual addition in $\mathbb{R}$.
(B)	V is a subspace of the real vector space $\mathbb{R}$.
(C)	V is a two dimensional subspace of the vector space $\mathbb{R}$ over $\mathbb{Q}$.
(D)	V is a four dimensional subspace of the vector space $\mathbb{R}$ over $\mathbb{Q}$.

Section C: Q.41 – Q.50 Carry ONE mark each.

Question 2 (JAM 2026 • Q28 • ID: JAM_2026_Q28) MCQ +2 M Neg: −0.67

Q.28	For any two points $P, Q \in \mathbb{R}^3$, let $vect(P,Q)$ denote the vector from the point P to the point Q and let $d(P,Q)$ denote the length of $vect(P,Q)$. Let $F: \mathbb{R}^3 \longrightarrow \mathbb{R}^3$ be a linear transformation such that $$d\big(F(P), F(Q)\big) = d(P,Q) \quad \text{for all } P, Q \in \mathbb{R}^3.$$ Which ONE of the following statements is FALSE ?
(A)	F is an injective map.
(B)	F is a surjective map.
(C)	For any four points P, Q, R, S in $\mathbb{R}^3$, we have $vect\big(F(P), F(Q)\big) \cdot vect\big(F(R), F(S)\big) = vect(P,Q) \cdot vect(R,S)$, where $\cdot$ denotes the usual dot product in $\mathbb{R}^3$.
(D)	For any four points P, Q, R, S in $\mathbb{R}^3$, we have $vect\big(F(P), F(Q)\big) \times vect\big(F(R), F(S)\big) = vect(P,Q) \times vect(R,S)$, where $\times$ denotes the usual cross product in $\mathbb{R}^3$.

Question 3 (JAM 2026 • Q24 • ID: JAM_2026_Q24) **MCQ** **+2 M** **Neg: −0.67**

Q.24	Let e_1, e_2, e_3, e_4, e_5 denote the standard basis vectors of the real vector space $\mathbb{R}^5$. Let V be the subspace of $\mathbb{R}^5$ spanned by the vectors $e_1 + e_2, e_2 + e_3, e_3 + e_4 + e_5$. Consider the real vector space $M_{2,5}(\mathbb{R})$. Let $$S = \{P \in M_{2,5}(\mathbb{R}) \mid \text{nullspace}(P) = V\}.$$ Which ONE of the following statements is TRUE?
(A)	S is the empty set.
(B)	$S = \{0\}$.
(C)	S is non-empty and S is not a subspace of $M_{2,5}(\mathbb{R})$.
(D)	S is a nonzero subspace of $M_{2,5}(\mathbb{R})$.

Question 4 (JAM 2025 • Q58 • ID: JAM_2025_Q58) **NAT** **+2 M** **Neg: Nil (0)**

Q.58 Let the subspace H of $P_3(\mathbb{R})$ be defined as

$$H = \{p(x) \in P_3(\mathbb{R}) : xp'(x) = 3p(x)\}.$$

Then, the dimension of H is equal to ________________.

Question 5 (JAM 2025 • Q49 • ID: JAM_2025_Q49) **NAT** **+1 M** **Neg: Nil (0)**

Q.49 Let $T, S : P_4(\mathbb{R}) \to P_4(\mathbb{R})$ be the linear transformations defined by

$$T(p(x)) = xp'(x) \text{ and } S(p(x)) = (x+1)p'(x)$$

for all $p(x) \in P_4(\mathbb{R})$.

Then, the nullity of the composition $S \circ T$ is ____________.

Question 6 (JAM 2025 • Q47 • ID: JAM_2025_Q47) **NAT** **+1 M** **Neg: Nil (0)**

Q.47 Consider the real vector space $\mathbb{R}^3$. Let $T : \mathbb{R}^3 \to \mathbb{R}$ be a linear transformation such that

$$T(1,1,1) = 0, \qquad T(1,-1,1) = 0 \text{ and } T(0,0,1) = 16.$$

Then, the value of $T\left(\frac{1}{2}, \frac{2}{3}, \frac{3}{4}\right)$ is equal to __________________ *(rounded off to two decimal places).*

Question 7 (JAM 2025 • Q46 • ID: JAM_2025_Q46) **NAT** **+1 M** **Neg: Nil (0)**

Q.46 Consider the following subspaces of $\mathbb{R}^4$:

$$V_1 = \{(x,y,z,w) \in \mathbb{R}^4 : x + y + 2w = 0\},$$

$$V_2 = \{(x,y,z,w) \in \mathbb{R}^4 : 2y + z + w = 0\},$$

$$V_3 = \{(x,y,z,w) \in \mathbb{R}^4 : x + 3y + z + 3w = 0\}.$$

Then, the dimension of the subspace $V_1 \cap V_2 \cap V_3$ is equal to __________.

Question 8 (JAM 2025 • Q40 • ID: JAM_2025_Q40) MSQ +2 M Neg: Nil (0)

Q.40 Consider the following subspaces of the real vector space $\mathbb{R}^3$:

$V_1 = \text{span}\,\{(1,2,3),(1,1,0)\}$,

$V_2 = \text{span}\,\{(1,-1,0)\}$,

$V_3 = \text{span}\,\{(1,1,1)\}$,

$V_4 = \text{span}\,\{(1,3,6)\}$ and

$V_5 = \text{span}\,\{(1,0,-3)\}$.

Then, which of the following is/are TRUE?

(A) $V_1 \cup V_2$ is a subspace of $\mathbb{R}^3$

(B) $V_1 \cup V_3$ is a subspace of $\mathbb{R}^3$

(C) $V_1 \cup V_4$ is a subspace of $\mathbb{R}^3$

(D) $V_1 \cup V_5$ is a subspace of $\mathbb{R}^3$

Question 9 (JAM 2025 • Q35 • ID: JAM_2025_Q35) MSQ +2 M Neg: Nil (0)

Q.35 Let

$u_1 = (1,0,0,-1), u_2 = (0,2,0,-1),\ u_3 = (0,0,1,-1)$ and $u_4 = (0,0,0,1)$

be elements in the real vector space $\mathbb{R}^4$.

Then, which of the following is/are TRUE?

(A) $\{u_1,\ u_2,\ u_3,\ u_4\}$ is a linearly independent set in $\mathbb{R}^4$

(B) $\{u_1 - u_2, u_2 - u_3,\ u_3 - u_4, u_4 - u_1\}$ is NOT a linearly independent set in $\mathbb{R}^4$

(C) $\{u_1,\ -u_2,\ u_3, -u_4\}$ is NOT a linearly independent set in $\mathbb{R}^4$

(D) $\{u_1 + u_2,\ u_2 +\ u_3,\ u_3 + u_4,\ u_4 + u_1\}$ is a linearly independent set in $\mathbb{R}^4$

Question 10 (JAM 2025 • Q24 • ID: JAM_2025_Q24) **MCQ** **+2 M** **Neg: −0.67**

Q.24 Let f_1, f_2, f_3 be nonzero linear transformations from $\mathbb{R}^4$ to $\mathbb{R}$ and

$$\ker(f_1) \subseteq \ker(f_2) \cap \ker(f_3).$$

Let $T : \mathbb{R}^4 \to \mathbb{R}^3$ be the linear transformation defined by

$$T(v) = \big(f_1(v), f_2(v), f_3(v)\big), \qquad \text{for all } v \in \mathbb{R}^4.$$

Then, the nullity of T is equal to

(A) 1

(B) 2

(C) 3

(D) 4

Question 11 (JAM 2025 • Q8 • ID: JAM_2025_Q8) MCQ +1 M Neg: −0.33

Q.8 Let $\mathrm{T} : P_2(\mathbb{R}) \to P_2(\mathbb{R})$ be the linear transformation defined by $\mathrm{T}\big(p(x)\big) = p(x+1)$, for all $p(x) \in P_2(\mathbb{R})$. If M is the matrix representation of T with respect to the ordered basis $\{1, x, x^2\}$ of $P_2(\mathbb{R})$, then which one of the following is TRUE?

(A) The determinant of M is 2

(B) The rank of M is 2

(C) 1 is the only eigenvalue of M

(D) The nullity of M is 2

Question 12 (JAM 2025 • Q6 • ID: JAM_2025_Q6) MCQ +1 M Neg: −0.33

Q.6 Define $\mathrm{T}: \mathbb{R}^3 \to \mathbb{R}^3$ by

$$\mathrm{T}(x, y, z) = (x + z,\ 2x + 3y + 5z,\ 2y + 2z), \text{ for all } (x, y, z) \in \mathbb{R}^3.$$

Then, which one of the following is TRUE?

(A) T is one-one and T is NOT onto

(B) T is NOT one-one and T is onto

(C) T is one-one and T is onto

(D) T is NOT one-one and T is NOT onto

Question 13 (JAM 2024 • Q58 • ID: JAM_2024_Q58) NAT +2 M Neg: Nil (0)

Q.58 For $k \in \mathbb{N}$, let $0 = t_0 < t_1 < \cdots < t_k < t_{k+1} = 1$. A function $f: [0,1] \to \mathbb{R}$ is said to be piecewise linear with nodes $t_1, \dots, t_k$, if for each $j = 1, 2, \dots, k+1$, there exist $a_j \in \mathbb{R}$, $b_j \in \mathbb{R}$ such that

$$f(t) = a_j + b_j t \text{ for } t_{j-1} < t < t_j .$$

Let V be the real vector space of all real valued continuous piecewise linear functions on $[0,1]$ with nodes $\frac{1}{4}, \frac{1}{2}, \frac{3}{4}$. Then, the dimension of V equals

____

Question 14 (JAM 2024 • Q49 • ID: JAM_2024_Q49) NAT +1 M Neg: Nil (0)

Q.49 Let $P_7(x)$ be the real vector space of polynomials in x with degree at most 7, together with the zero polynomial. For $r = 1, 2, \dots, 7$, define

$$s_r(x) = x(x-1)\cdots\big(x-(r-1)\big) \text{ and } s_0(x) = 1.$$

Consider the fact that $B = \{s_0(x), s_1(x), \dots, s_7(x)\}$ is a basis of $P_7(x)$.

If

$$x^5 = \sum_{k=0}^{7} \alpha_{5,k}\, s_k(x) ,$$

where $\alpha_{5,k} \in \mathbb{R}$, then $\alpha_{5,2}$ equals ____________ (*rounded off to two decimal places*)

Question 15 (JAM 2024 • Q45 • ID: JAM_2024_Q45) NAT +1 M Neg: Nil (0)

Q.45 Let $P_{12}(x)$ be the real vector space of polynomials in the variable x with real coefficients and having degree at most 12, together with the zero polynomial. Define

$$V = \left\{ f \in P_{12}(x)\colon\ f(-x) = f(x) \text{ for all } x \in \mathbb{R} \text{ and } f(2024) = 0 \right\}.$$

Then, the dimension of V is ______________

Question 16 (JAM 2024 • Q22 • ID: JAM_2024_Q22) MCQ +2 M Neg: −0.67

Q.22 Let $P_7(x)$ be the real vector space of polynomials, in the variable x with real coefficients and having degree at most 7, together with the zero polynomial. Let $T\colon\ P_7(x) \to P_7(x)$ be the linear transformation defined by

$$T\big(f(x)\big) = f(x) + \frac{df(x)}{dx}\,.$$

Then, which one of the following is TRUE?

(A) T is not a surjective linear transformation

(B) There exists $k \in \mathbb{N}$ such that T^k is the zero linear transformation

(C) 1 and 2 are the eigenvalues of T

(D) There exists $r \in \mathbb{N}$ such that $(T - I)^r$ is the zero linear transformation, where I is the identity map on $P_7(x)$

Question 17 (JAM 2024 • Q17 • ID: JAM_2024_Q17) MCQ +2 M Neg: −0.67

Q.17 Let $P_{11}(x)$ be the real vector space of polynomials, in the variable x with real coefficients and having degree at most 11, together with the zero polynomial. Let

$$E = \{s_0(x), s_1(x), \dots, s_{11}(x)\}, \qquad F = \{r_0(x), r_1(x), \dots, r_{11}(x)\}$$

be subsets of $P_{11}(x)$ having 12 elements each and satisfying

$$s_0(3) = s_1(3) = \cdots = s_{11}(3) = 0\,, \qquad r_0(4) = r_1(4) = \cdots = r_{11}(4) = 1\,.$$

Then, which one of the following is TRUE?

(A) Any such E is not necessarily linearly dependent and any such F is not necessarily linearly dependent

(B) Any such E is necessarily linearly dependent but any such F is not necessarily linearly dependent

(C) Any such E is not necessarily linearly dependent but any such F is necessarily linearly dependent

(D) Any such E is necessarily linearly dependent and any such F is necessarily linearly dependent

Question 18 (JAM 2024 • Q13 • ID: JAM_2024_Q13) MCQ +2 M Neg: −0.67

Q.13 Let V be a nonzero subspace of the complex vector space $M_7(\mathbb{C})$ such that every nonzero matrix in V is invertible. Then, the dimension of V over $\mathbb{C}$ is

(A) 1

(B) 2

(C) 7

(D) 49

Question 19 (JAM 2023 • Q57 • ID: JAM_2023_Q57) NAT +2 M Neg: Nil (0)

Q. 57 Let W be the subspace of $M_3(\mathbb{R})$ consisting of all matrices with the property that the sum of the entries in each row is zero and the sum of the entries in each column is zero. Then the dimension of W is equal to ___________.

Question 20 (JAM 2023 • Q42 • ID: JAM_2023_Q42) NAT +1 M Neg: Nil (0)

Q. 42 Let $T : P_2(\mathbb{R}) \to P_4(\mathbb{R})$ be the linear transformation given by $T(p(x)) = p(x^2)$. Then the rank of T is equal to ___________.

Question 21 (JAM 2023 • Q39 • ID: JAM_2023_Q39) **MSQ** **+2 M** **Neg: Nil (0)**

Q. 39 Which of the following is/are linear transformations?

(A) $T : \mathbb{R} \to \mathbb{R}$ given by $T(x) = \sin(x)$

(B) $T : M_2(\mathbb{R}) \to \mathbb{R}$ given by $T(A) = \text{trace}(A)$

(C) $T : \mathbb{R}^2 \to \mathbb{R}$ given by $T(x, y) = x + y + 1$

(D) $T : P_2(\mathbb{R}) \to \mathbb{R}$ given by $T(p(x)) = p(1)$

Question 22 (JAM 2023 • Q38 • ID: JAM_2023_Q38) **MSQ** **+2 M** **Neg: Nil (0)**

Q. 38 Which of the following is/are true?

(A) Every linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^2$ maps lines onto points or lines

(B) Every surjective linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^2$ maps lines onto lines

(C) Every bijective linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^2$ maps pairs of parallel lines to pairs of parallel lines

(D) Every bijective linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^2$ maps pairs of perpendicular lines to pairs of perpendicular lines

Question 23 (JAM 2023 • Q15 • ID: JAM_2023_Q15) MCQ +2 M Neg: −0.67

Q. 15 Let S and T be non-empty subsets of $\mathbb{R}^2$, and W be a non-zero proper subspace of $\mathbb{R}^2$. Consider the following statements:

I. If $\text{span}(S) = \mathbb{R}^2$, then $\text{span}(S \cap W) = W$.

II. $\text{span}(S \cup T) = \text{span}(S) \cup \text{span}(T)$.

Then

(A) both I and II are TRUE

(B) I is TRUE but II is FALSE

(C) II is TRUE but I is FALSE

(D) both I and II are FALSE

Question 24 (JAM 2023 • Q12 • ID: JAM_2023_Q12) MCQ +2 M Neg: −0.67

Q. 12 Consider the following statements:

I. There exists a linear transformation from $\mathbb{R}^3$ to itself such that its range space and null space are the same.

II. There exists a linear transformation from $\mathbb{R}^2$ to itself such that its range space and null space are the same.

Then

(A) both I and II are TRUE

(B) I is TRUE but II is FALSE

(C) II is TRUE but I is FALSE

(D) both I and II are FALSE

Question 25 (JAM 2023 • Q3 • ID: JAM_2023_Q3) MCQ +1 M Neg: −0.33

Q. 3 Which of the following is a subspace of the real vector space $\mathbb{R}^3$?

(A) $\{(x, y, z) \in \mathbb{R}^3 : (y + z)^2 + (2x - 3y)^2 = 0\}$

(B) $\{(x, y, z) \in \mathbb{R}^3 : y \in \mathbb{Q}\}$

(C) $\{(x, y, z) \in \mathbb{R}^3 : yz = 0\}$

(D) $\{(x, y, z) \in \mathbb{R}^3 : x + 2y - 3z + 1 = 0\}$

Question 26 (JAM 2022 • Q40 • ID: JAM_2022_Q40) MSQ +2 M Neg: Nil (0)

Q.40	Let P be a fixed 3×3 matrix with entries in $\mathbb{R}$. Which of the following maps from $M_3(\mathbb{R})$ to $M_3(\mathbb{R})$ is/are linear?
(A)	$T_1: M_3(\mathbb{R}) \to M_3(\mathbb{R})$ given by $T_1(M) = MP - PM$ for $M \in M_3(\mathbb{R})$.
(B)	$T_2: M_3(\mathbb{R}) \to M_3(\mathbb{R})$ given by $T_2(M) = M^2P - P^2M$ for $M \in M_3(\mathbb{R})$.
(C)	$T_3: M_3(\mathbb{R}) \to M_3(\mathbb{R})$ given by $T_3(M) = MP^2 + P^2M$ for $M \in M_3(\mathbb{R})$.
(D)	$T_4: M_3(\mathbb{R}) \to M_3(\mathbb{R})$ given by $T_4(M) = MP^2 - PM^2$ for $M \in M_3(\mathbb{R})$.
Section C: Q.41 – Q.50 Carry ONE mark each.	

Question 27 (JAM 2022 • Q39 • ID: JAM_2022_Q39) MSQ +2 M Neg: Nil (0)

Q.39	Let V be the real vector space consisting of all polynomials in one variable with real coefficients and having degree at most 5, together with the zero polynomial. Let $T: V \to \mathbb{R}$ be the linear map defined by $T(1) = 1$ and $$T\big(x(x-1)\cdots(x-k+1)\big) = 1 \quad \text{for } 1 \le k \le 5.$$ Then which of the following is/are true?
(A)	$T(x^4) = 15$.
(B)	$T(x^3) = 5$.
(C)	$T(x^4) = 14$.
(D)	$T(x^3) = 3$.

Question 28 (JAM 2022 • Q3 • ID: JAM_2022_Q3) MCQ +1 M Neg: −0.33

Q.3	Let V be the real vector space consisting of all polynomials in one variable with real coefficients and having degree at most 6, together with the zero polynomial. Then which one of the following is true?
(A)	$\{f \in V : f(1/2) \notin \mathbb{Q}\}$ is a subspace of V.
(B)	$\{f \in V : f(1/2) = 1\}$ is a subspace of V.
(C)	$\{f \in V : f(1/2) = f(1)\}$ is a subspace of V.
(D)	$\{f \in V : f'(1/2) = 1\}$ is a subspace of V.

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Finite Dimensional Vector Spaces (2022–2026) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	MSQ	+2	**B;D**
2	MCQ	+2	**D**
3	MCQ	+2	**C**
4	NAT	+2	**1 to 1**
5	NAT	+1	**1 to 1**
6	NAT	+1	**3.98 to 4.02**
7	NAT	+1	**2 to 2**
8	MSQ	+2	**C;D**
9	MSQ	+2	**A;B**
10	MCQ	+2	**C**
11	MCQ	+1	**C**
12	MCQ	+1	**D**
13	NAT	+2	**5 to 5**
14	NAT	+1	**14.95 to 15.05**

Q. No.	Type	Marks	Official Key
15	NAT	+1	**6 to 6**
16	MCQ	+2	**D**
17	MCQ	+2	**B**
18	MCQ	+2	**A**
19	NAT	+2	**4 to 4**
20	NAT	+1	**3 to 3**
21	MSQ	+2	**B, D**
22	MSQ	+2	**A, B, C**
23	MCQ	+2	**D**
24	MCQ	+2	**C**
25	MCQ	+1	**A**
26	MSQ	+2	**A,C**
27	MSQ	+2	**A,B**
28	MCQ	+1	**C**

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt (-1/3 for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., [a, b]) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.